SUNLIGHT

Verification of Five Procurement Contracts under the v4.0 Core Architecture:
Findings, Methodology, and Reproducibility Notes

Le present document rapporte les resultats de verification de cinq contrats d'approvisionnement a travers l'architecture SUNLIGHT v4.0 Core. Chaque contrat est soumis aux trois moteurs analytiques du systeme — l'indicateur statistique de risque concurrentiel (CRI), l'analyse topologique contractuelle (TCA) et la porte de verification des preuves (EVG) — produisant un verdict a trois niveaux (GREEN, YELLOW, RED) avec tracabilite complete des regles appliquees, des seuils franchis et des citations juridiques rendues par le profil juridictionnel applicable. Les cinq cas couvrent le spectre analytique complet : absence de constat, anomalie statistique unidimensionnelle, constat structurel multidimensionnel, validation de reference sur un cas poursuivi par le DOJ americain, et comparaison de comportement entre deux profils juridictionnels sur un contrat identique.

The technical body of this document presents each contract as a self-contained case memorandum. All numerical values, thresholds, verdicts, and legal citations are drawn directly from the engine output recorded in the analytical register (contracts_analyzed.json) generated by the SUNLIGHT v4.0 Core pipeline. No value in this document has been rounded, edited, or interpreted beyond what the engine produced. A methodology appendix follows the five case memoranda.

Document Metadata

Engine version	v4_core
HEAD commit	244f991
MJPIS markup floor ratio	0.501
MJPIS bribery channel ratio	0.0058
MJPIS sanctionable threshold	18 months

Test suite passing	682
DOJ regression baseline	69.2% / 100% / 8.0% / TP=9 FP=4 FN=0 TN=46
Generation timestamp	2026-04-14T13:57:59.702495+00:00

Table of Findings

Case	Contract ID	Agency / Vendor	Profile	Verdict	Dims Firing
1	WE31	Dept. of Defense / TETRA TECH INC	us_federal	GREEN	0
2	75N91020F00014	HHS / LEIDOS BIOMEDICAL RESEARCH	us_federal	YELLOW	1
3	N0002417C2117	Dept. of Defense / ELECTRIC BOAT CORP	us_federal	RED	2
4	US_v_Boeing_2006	Dept. of Defense / Boeing Company	us_federal	RED	2
5a	W9113M08C0031	Dept. of Defense / CONCURRENT TECH	us_federal	YELLOW	1
5b	W9113M08C0031	Dept. of Defense / CONCURRENT TECH	uk_central_gov	RED	2

Case 1: GREEN Baseline

I. Contract Identification

Awarding agency	Department of Defense
Vendor	TETRA TECH INC
Contract value	USD 3,080,692
Award date	2017-05-26
Description	IGF::OT::IGF NWS EARLE FY17 BASEWIDE SUPPORT
Procurement method	open

II. Verdict

The Evidence Verification Gate returned a GREEN verdict. No evidence dimension exceeded its applicable threshold under the us_federal_v0 parameter set.

III. Evidence Dimensions

Dimension	Observed Value	Threshold	Status
CRI Markup Ratio	-0.8759	0.7500	Below threshold
CRI Bribery Channel	N/A	N/A	Below threshold
TCA Typologies	0	2	Below threshold

IV. Structural Findings

No structural contradiction rules fired under this profile. This case establishes the analytical baseline: a contract with no structural risk indicators under any of the three engine dimensions.

V. Reproducibility

Analysis timestamp: 2026-04-14T13:57:59.702495+00:00. Processing time: 1817.9 ms. Full ContractDossier available in demo/contracts_analyzed.json.

Case 2: YELLOW Single-Dimension Finding

I. Contract Identification

Awarding agency	Department of Health and Human Services
Vendor	LEIDOS BIOMEDICAL RESEARCH INC
Contract value	USD 50,000,000
Award date	2020-06-30
Description	COVID ACTIV3 TASK ORDER SUPPORTS NIAID'S GOAL TO BETTER UNDERSTAND THE CORONAVIRUS WITH AN INITIAL FOCUS ON ACTIV3. THIS TASK ORDER WILL INITIATE THE MANAGEMENT , OVERSIGHT , AND CONDUCT OF THE NIAID DCR CLINICAL TRIAL TITLED A MULTICENTER, ADA
Procurement method	open

II. Verdict

The Evidence Verification Gate returned a YELLOW verdict. One evidence dimension exceeded its applicable threshold under the us_federal_v0 parameter set, indicating a statistical anomaly warranting investigator review. A YELLOW verdict is not an allegation; it is a structured risk indicator.

III. Evidence Dimensions

Dimension	Observed Value	Threshold	Status
CRI Markup Ratio	1.1092	0.7500	Above threshold
CRI Bribery Channel	N/A	N/A	Below threshold
TCA Typologies	1	2	Below threshold

IV. Structural Findings

FIN-001: Award (USD 50,000,000) exceeds tender (USD 23,705,670) by 110.9%

> **Legal citations:** World Bank Procurement Framework; UNDP POPP Contract Modifications policy; MJPIS empirical markup floor (sub-task 2.3.7)

No TCA rules produced contradiction edges under this profile; the YELLOW verdict is driven entirely by the statistical CRI dimension.

V. Reproducibility

Analysis timestamp: 2026-04-14T13:57:59.702495+00:00. Processing time: 134.6 ms. Full ContractDossier available in demo/contracts_analyzed.json. CRI markup ratio: 1.1092 (110.92%). Bootstrap 95% CI: [103.4%, 117.8%]. Bayesian posterior: 0.666. Comparable contracts in peer group: 2312.

Case 3: RED Multi-Dimensional Finding (Production Data)

I. Contract Identification

Awarding agency	Department of Defense
Vendor	ELECTRIC BOAT CORPORATION
Contract value	USD 24,313,811,834
Award date	2017-09-21
Description	COLUMBIA CLASS DESIGN COMPLETION
Procurement method	open

II. Verdict

The Evidence Verification Gate returned a RED verdict. 2 evidence dimensions exceeded their applicable thresholds simultaneously under the us_federal_v0 parameter set. This RED verdict is produced on a contract from the production database with no prior prosecution history. SUNLIGHT surfaces the structural risk indicator; determination of intent and case disposition remains with institutional investigators.

III. Evidence Dimensions

Dimension	Observed Value	Threshold	Status
CRI Markup Ratio	11.5652	0.7500	Above threshold
CRI Bribery Channel	N/A	N/A	Below threshold
TCA Typologies	2	2	Above threshold

IV. Structural Findings

FIN-001: Award (USD 24,313,811,834) exceeds tender (USD 1,935,011,925) by 1156.5%

> **Legal citations:** World Bank Procurement Framework; UNDP POPP Contract Modifications policy; MJPIS empirical markup floor (sub-task 2.3.7)

TIME-001: Award in final 2 weeks of fiscal period (month 9, day 21)

> **Legal citations:** Fazekas fiscal pressure research; OECD public procurement timing analysis; Senegal case (award 8 days before fiscal close)

V. Reproducibility

Analysis timestamp: 2026-04-14T13:57:59.702495+00:00. Processing time: 38.7 ms. Full ContractDossier available in demo/contracts_analyzed.json. EVG methodology: EVG v1.0: GREEN=0 dims fired, YELLOW=1 dim, RED=2+ dims. Thresholds from us_federal_v0.

Case 4: DOJ-Prosecuted Reference Case (Credibility Floor)

I. Contract Identification

Awarding agency	Department of Defense
Vendor	Boeing Company
Contract value	USD 148,287,618
Description	Charged Air Force 300-600% markup on standard aircraft spare parts with known commercial prices
Procurement method	direct

II. Verdict

The Evidence Verification Gate returned a RED verdict. 2 evidence dimensions exceeded their applicable thresholds simultaneously under the us_federal_v0 parameter set. Documented fraud type: Spare Parts Overcharging. Documented markup: 450%. Settlement: $615,000,000.

III. Evidence Dimensions

Dimension	Observed Value	Threshold	Status
CRI Markup Ratio	4.3268	0.7500	Above threshold
CRI Bribery Channel	N/A	N/A	Below threshold
TCA Typologies	2	2	Above threshold

IV. Structural Findings

PROC-001: Direct award of USD 27,838,030 exceeds competitive threshold (USD 100,000)

> **Legal citations:** UNCAC Art. 9(1) — Public procurement systems based on transparency, competition and objective criteria; UNCAC Art. 9(2) — Public finance management measures; UNCAC Art. 12 — Private sector measures against corruption; UNCAC Art. 15 — Bribery of national public officials; UNCAC Art. 16 — Bribery of foreign public officials and officials of public international organizations; UNCAC Art. 17 — Embezzlement, misappropriation or other diversion of property by a public official; UNCAC Art. 18 — Trading in influence; OECD Anti-Bribery Convention 1997 — Convention on Combating Bribery of Foreign Public Officials in International Business Transactions; OECD Recommendation on Public Procurement 2015 — OECD/LEGAL/0411; Federal Acquisition Regulation (FAR) Part 6 — Competition Requirements; FAR Part 15 — Contracting by Negotiation; FAR Part 13 — Simplified Acquisition Procedures

FIN-001: Award (USD 148,287,618) exceeds tender (USD 27,838,030) by 432.7%

> **Legal citations:** World Bank Procurement Framework; UNDP POPP Contract Modifications policy; MJPIS empirical markup floor (sub-task 2.3.7)

V. Reproducibility

Analysis timestamp: 2026-04-14T13:57:59.702495+00:00. Processing time: 50.1 ms. Full ContractDossier available in demo/contracts_analyzed.json. This contract is among the nine DOJ-prosecuted reference cases that constitute SUNLIGHT's institutional credibility floor. Its RED verdict is preserved across every engine modification since the validation baseline was established; the 100% recall invariant on the nine cases holds byte-identically across more than thirty consecutive commits in the current architecture arc.

Case 5a: Jurisdiction Comparison (US Federal Profile)

I. Contract Identification

Awarding agency	Department of Defense
Vendor	CONCURRENT TECHNOLOGIES CORP
Contract value	USD 49,985,204
Award date	2008-03-24
Description	BASIC EFFORT - EXERCISED PORTION
Procurement method	open

II. Verdict

The Evidence Verification Gate returned a YELLOW verdict. One evidence dimension exceeded its applicable threshold under the us_federal_v0 parameter set, indicating a statistical anomaly warranting investigator review. A YELLOW verdict is not an allegation; it is a structured risk indicator.

III. Evidence Dimensions

Dimension	Observed Value	Threshold	Status
CRI Markup Ratio	0.8187	0.7500	Above threshold
CRI Bribery Channel	N/A	N/A	Below threshold
TCA Typologies	1	2	Below threshold

IV. Structural Findings

FIN-001: Award (USD 49,985,204) exceeds tender (USD 27,484,029) by 81.9%

> **Legal citations:** World Bank Procurement Framework; UNDP POPP Contract Modifications policy; MJPIS empirical markup floor (sub-task 2.3.7)

V. Reproducibility

Analysis timestamp: 2026-04-14T13:57:59.702495+00:00. Processing time: 2020.7 ms. Full ContractDossier available in demo/contracts_analyzed.json.

Case 5b: Jurisdiction Comparison (UK Central Government Profile)

I. Contract Identification

Awarding agency	Department of Defense
Vendor	CONCURRENT TECHNOLOGIES CORP
Contract value	USD 49,985,204
Award date	2008-03-24
Description	BASIC EFFORT - EXERCISED PORTION
Procurement method	open

II. Verdict

The Evidence Verification Gate returned a RED verdict. 2 evidence dimensions exceeded their applicable thresholds simultaneously under the uk_central_government parameter set.

III. Evidence Dimensions

Dimension	Observed Value	Threshold	Status
CRI Markup Ratio	0.8187	0.7500	Above threshold
CRI Bribery Channel	N/A	N/A	Below threshold
TCA Typologies	2	2	Above threshold

IV. Structural Findings

FIN-001: Award (USD 49,985,204) exceeds tender (USD 27,484,029) by 81.9%

> **Legal citations:** World Bank Procurement Framework; UNDP POPP Contract Modifications policy; MJPIS empirical markup floor (sub-task 2.3.7)

TIME-001: Award in final 2 weeks of fiscal period (month 3, day 24)

> **Legal citations:** Fazekas fiscal pressure research; OECD public procurement timing analysis; Senegal case (award 8 days before fiscal close)

V. Reproducibility

Analysis timestamp: 2026-04-14T13:57:59.702495+00:00. Processing time: 2023.3 ms. Full ContractDossier available in demo/contracts_analyzed.json.

Jurisdiction Delta Commentary

The behavioral delta between the two profiles is not stylistic but structural. TIME-001 fires under UK because the contract's award date (2008-03-24) falls within seven days of

the UK fiscal year-end (March 31); it does not fire under US federal because March is a safe month in the US federal fiscal calendar (year-end September 30). The same contract code, the same engine, the same 16 TCA rules — two jurisdiction profiles, two jurisdiction-correct outputs, no code change. This is the JurisdictionProfile architecture in operational form.

Explicit Deltas

Rules firing only under UK profile: ['TIME-001']

Structural confidence: US=0.75, UK=0.55

EVG verdict: US=yellow, UK=red

Legal citation deltas: 1 rules have different citations

Appendix A — Analytical Methodology and Reproducibility

Three-Engine Architecture

SUNLIGHT employs three analytically independent engines operating in sequence on each contract. The Competitive Risk Indicator (CRI) engine performs statistical price analysis, comparing the contract's award value against a peer group of comparable contracts derived from the production database, producing bootstrap confidence intervals and Bayesian posterior probabilities for markup significance. The Topological Contract Analysis (TCA) engine constructs a directed graph representing the contract's structural relationships — procurement method, financial flows, timing, geographic jurisdiction, entity ownership — and evaluates 16 rules drawn from international anti-corruption frameworks, producing contradiction (REMOVES), aspiration (SEEKS), and verification (VERIFIES) edges. The Evidence Verification Gate (EVG) combines the outputs of both engines against calibrated thresholds to produce the final tiered verdict.

MJPIS Intersection Methodology

The Minimum Jurisdictional Prosecutorial Intersection Standard (MJPIS) derives analytical thresholds from a 32-case corpus spanning four mature legal systems: US Department of Justice, UK Serious Fraud Office, French Parquet National Financier, and World Bank Integrity Vice Presidency. For each analytical parameter, the MJPIS derivation identifies the value that would survive evidentiary scrutiny in all four jurisdictions simultaneously. Current derived values: markup floor ratio 0.501, bribery-channel ratio 0.0058, administrative sanctionable threshold 18 months. This intersection methodology produces thresholds that are architecturally over-defensible rather than under-defensible.

Jurisdiction Profile Architecture

SUNLIGHT implements jurisdiction-specific behavior through parameterized profiles rather than jurisdiction-specific code. Four operational profiles are currently deployed: us_federal, uk_central_government, fr_central_government, and wb_international. Each profile specifies fiscal calendar parameters, competitive thresholds, entity registration requirements, and legal citation libraries. Adding a new country profile is a data parameterization task, not a code change. The same 16 TCA rules and the same EVG gate logic operate identically across all profiles; the behavioral delta between jurisdictions is produced entirely by the profile parameters.

Evidence Verification Gate and Verdict Tiering

The EVG evaluates three evidence dimensions: CRI Markup Ratio (observed markup against the MJPIS floor), CRI Bribery Channel (reserved for future implementation), and TCA Typologies (count of distinct structural contradiction rules fired). The verdict tiers are: GREEN (zero dimensions above threshold), YELLOW (exactly one dimension above

threshold), and RED (two or more dimensions above threshold simultaneously). This multi-dimensional gating ensures that no single statistical anomaly or single structural finding can produce a RED verdict in isolation.

Test Suite and Regression Invariants

The SUNLIGHT codebase maintains 682 automated tests covering statistical correctness, structural rule evaluation, EVG gate logic, API endpoints, and security hardening. The institutional credibility floor is maintained through a 100% recall invariant on nine DOJ-prosecuted reference cases, verified byte-identically at every commit: Precision 33.3%, Recall 100.0%, FPR 9.0%, PR-AUC 0.746, Flags/1K 129.2. The EVG integration test (added at commit 85f6ed7) exercises the real production data flow from ContractDossier through TCAGraphRuleEngine, analyze_tca_graph, and the EVG gate, closing the key-mismatch bug class identified at commit 706df82 with counterfactual verification.